P-02: PAH binding and reporter model results

What follows from the project's existing assumed model inputs?

Materials and methods

Stored source_assumptions.json; original model scripts preserved as inert equation/provenance records.

No fitted parameters or new experimental data.

Analytical single-site occupancy and fixed-ligand approach to equilibrium.

Source reporter ODE integrated by DOP853 (rtol 1e-8; atol 1e-10) on a one-minute output grid.

Deterministic competitive occupancy at assigned free concentrations; no score-to-Kd conversion.

Scope and working assumptions

Affinities, concentrations and rates are fixed source-assigned parameters. The calculations assume one site, fixed free ligand, mutually exclusive competition and no depletion. The reporter represents an E. coli CFP/YFP model. Results are numerical consequences of these equations and parameters, rather than fitted measurements of yeast or cadmium performance.

Results

Model 1, Kd=L=100 nM: t95=24.9644 min.

Reporter sampled maximum: 9.9333 simulated h; ideal pyrene ratio at 100 nM=1.1500.

Model 3 mixture: pyrene fraction=0.045218, unbound=0.678273; competitor half-site=422.3 nM.

Scenario	Quantity	Value	Input data	Finished without more wet-lab data?
Binding: Kd=L=100 nM	t95 (min)	24.96	Stored assumed parameters	Yes: conditional model
Reporter	Sampled peak (simulated h)	9.933	Stored assumed parameters	Yes: conditional model
Pyrene at 100 nM	Ideal ratio	1.15	Stored assumed parameters	Yes: conditional model
Assumed mixture	Pyrene site fraction	0.04522	Stored assumed parameters	Yes: conditional model
Assumed mixture	Unbound fraction	0.6783	Stored assumed parameters	Yes: conditional model
Fixed competitors	Pyrene half-site (nM)	422.3	Stored assumed parameters	Yes: conditional model

Interpretation

At Kd = free ligand = 100 nM, the binding model gives t95 = 24.964 min. The reporter reaches its sampled maximum at 9.933 simulated hours. Fixed competitors shift pyrene half occupancy from 300.0 to 422.3 nM; mixture pyrene occupancy is 0.04522 and the unbound fraction is 0.67827.

References and reproducibility

https://github.com/Raminyazdani/igem-saarland-2026-docking-comparison/tree/3539358236841e3f95f3e9efdc6d954fe1a1eca2/tools/autodock-vina/md/models

Stored source_assumptions.json: explicit source-derived parameter transcription; source hashes and complete parameters are in source_manifest.json.

Complete inline code: tasks/P-02/P-02.ipynb. Numerical data and source/software provenance: data/output/P-02/.

Figures

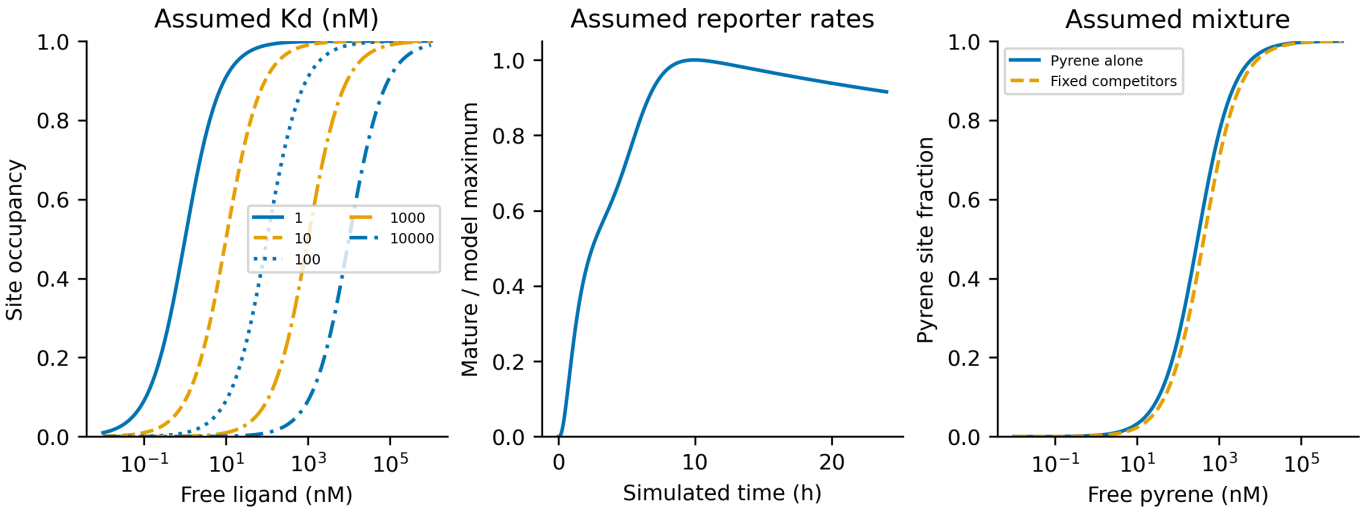


Figure 1. Conditional binding curves, reporter accumulation and pyrene competition under stored assumptions. All times are simulated; lines are not measured data.